Pranith Ravella

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Education

Purdue University West Lafayette, IN
Bachelor of Science in Computer Science *Expected May 2028*
Bachelor of Science in Mathematical Statistics

- **GPA**: 3.74 / 4.0
- **Activities/Clubs**: Boiler Quant Finance Group, Machine Learning @ Purdue, Sports Analytics @ Purdue
- **Relevant Coursework:** Operating Systems, Computer Networks, Data Structures & Algorithms, Computer Architecture, Systems Programming, C Programming, Linear Algebra, Discrete Math, Probability, Real Analysis

Experience

Software Engineer Intern May 2026 – August 2026
WellSky *Overland Park, KS*

- Built a Python automation platform replacing legacy UiPath automations, validated by 600 tests at 98% coverage
- Automated 200+ daily medical data entry and transfer jobs across hospital systems at a 95% success rate
- Developed a computer vision layer with OpenCV to extract text from UI screenshots to drive bot navigation
- Designed detection cascade caching algorithms to confirm element positions, dropping paid API calls by 80%

Software Engineer January 2026 – May 2026
Boiler Quant Finance Group *West Lafayette, IN*

- Programmed a CUDA options pricing engine using a parallelized CRR binomial tree, assigning one GPU block per contract and achieving 3.8M options/sec with a 480x throughput gain over CPU execution
- Wrote a parallel implied volatility solver using bisection search, attaining 100k+ solves/sec with sub-0.001% error
- Designed a two-kernel GPU pipeline for intra-tree and intra-option parallelism, maximizing GPU occupancy

Machine Learning Engineer August 2024 – May 2025
The Data Mine - Purdue University *West Lafayette, IN*

- Implemented a PyTorch object detection model to detect weeds in UAV imagery of farms, achieving 80% accuracy
- Engineered a Python script using Geopandas to automate the conversion of 100+ shapefiles containing weed detections into precise coordinate data, yielding a 95% accuracy in weed location
- Deployed the ML pipeline to the drone's software for edge inference, enabling execution of the detection model

Projects

CUDA LLM Inference Engine | *CUDA, C++, Python, PyTorch, Docker, Bash, AWS EC2*

- Built an LLM inference engine achieving 150 tokens/sec decode throughput, 2.7x faster than a PyTorch baseline
- Programmed a paged cache over token hashes, reducing KV cache block usage by 54% on shared prefix prompts
- Integrated a continuous batching scheduler with 2.4x throughput over static batching on an 80k ShareGPT trace

Open-Source Developments - Link

- **Reprobe:** Built a security tool to catch AI agents obeying hidden malicious instructions across sandboxed runs
- **Mandate:** Built a system to limit coding agents to only the access each task needs, cutting permissions by 99.8%

Cost Aware Kubernetes Scheduler - Link | *Go, Kubernetes, JavaScript, AWS (EC2 + EKS), Redis, Docker*

- Developed a cost aware scheduler for Amazon EKS, placing pods on the cheapest feasible node without violating SLOs, reducing compute cost by 18% and cross-AZ network traffic by 43–68% versus the default scheduler
- Designed a Go scoring plugin to rank nodes across five objectives, raising safe Spot utilization from 13% to 72%

Agentic Kafka DLQ Recovery - Link | *Go, Python, Kubernetes, ClickHouse, Kafka, Redis, Docker*

- Created a Kafka DLQ recovery platform to cluster failures and load 1M ClickHouse records at 296K events/second
- Engineered an AI investigator across 16 MCP tools to correlate Kafka schemas, deployments, and logs, achieving 92.9% root-cause accuracy, 100% repair accuracy, and zero unsafe recommendations

Technical Skills

Languages: Python, Go, C/C++, SQL, JavaScript, TypeScript
Frameworks: React, Node.js, Next.js, PyTorch, FastAPI, OpenTelemetry
Systems and Tools: CUDA, Linux, Kubernetes, Docker, ClickHouse, AWS, Redis, Kafka